

Battery Plan

Battery

Parameter	Value
Capacity	360 mAh
Voltage	7.4 V
Configuration	2S
Discharge rating	30C

Placement

The battery is mounted **above the drivetrain**, positioned **slightly toward the rear of the chassis**.

Why placement matters

The battery is one of the heavier single components in the vehicle. The design ideology treats the battery and the primary motor as a significant portion of the overall mass, which makes where the battery sits a mechanical decision rather than a packaging convenience.

Two properties of the vehicle make the position relevant:

- The front axle carries the Ackermann steering mechanism, so the load on the front wheels affects steering behaviour.
- The rear axle is a solid driven axle, so mass carried over it bears directly on the drivetrain.

The governing relationship is this: **the larger the mass bearing on the rear axle relative to the mass bearing on the Ackermann steered front axle, the less authority the steering system has, which results in severe understeer**. Steering authority depends on the front wheels being loaded enough to generate lateral force. As mass shifts rearward, that load falls, the front wheels lose their ability to turn the vehicle, and the vehicle continues straighter than the commanded steering angle.

Battery position is therefore constrained at both ends. It cannot sit far to the rear without unloading the steered axle, and it cannot sit far forward without demanding structure and space that the compact chassis does not have. Mounting it above the drivetrain places it low in the

chassis and over the region where the drive hardware is already supported, and the slight rearward offset keeps the package within the existing structure while staying close to the drivetrain it sits above.

Notes

This document records the battery specification and its position in the current design. Power distribution, regulation and wiring are documented in the electrical documentation.